# Monetary Policy

CHAPTER 15 · PRINCIPLES OF MACROECONOMICS

## Money demand, interest rates, and aggregate demand

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# Two FOMC Decisions

| Earlier decision cards | June 15, 2022 · Case C | September 18, 2024 · Case E |
|---|---|---|
| **Change in target range** | Increase of 0.75 percentage point | Decrease of 0.50 percentage point |
| **New target range** | 1.50–1.75% | 4.75–5.00% |

- The 2024 cut left the target range above its level after the 2022 increase.
- How could either decision change the money people hold and the purchases firms make?

FOMC, June 15, 2022 · FOMC, September 18, 2024

# Learning Objectives

- A · Understand and explain demand for money.
- B · Explain money-market equilibrium and the connection between money and interest rates.
- C · Interpret aggregate-demand shifts and demonstrate the effects of monetary policy.

# 15.1 Demand for Money

- Understand money demand as a desired stock for payments and liquidity.
- Explain how an investment’s return changes the opportunity cost of holding money.
- Distinguish a movement along money demand from a shift caused by other conditions.

# Money for Bills or Money for a Return?

- Use the separate PollEv activity: P1–P4. All amounts, rates, and contracts are hypothetical.
- Choose for the fictional household. Keep enough money for its upcoming bills.
- Keep your reasoning: we will use the first responses to build a money-demand comparison.

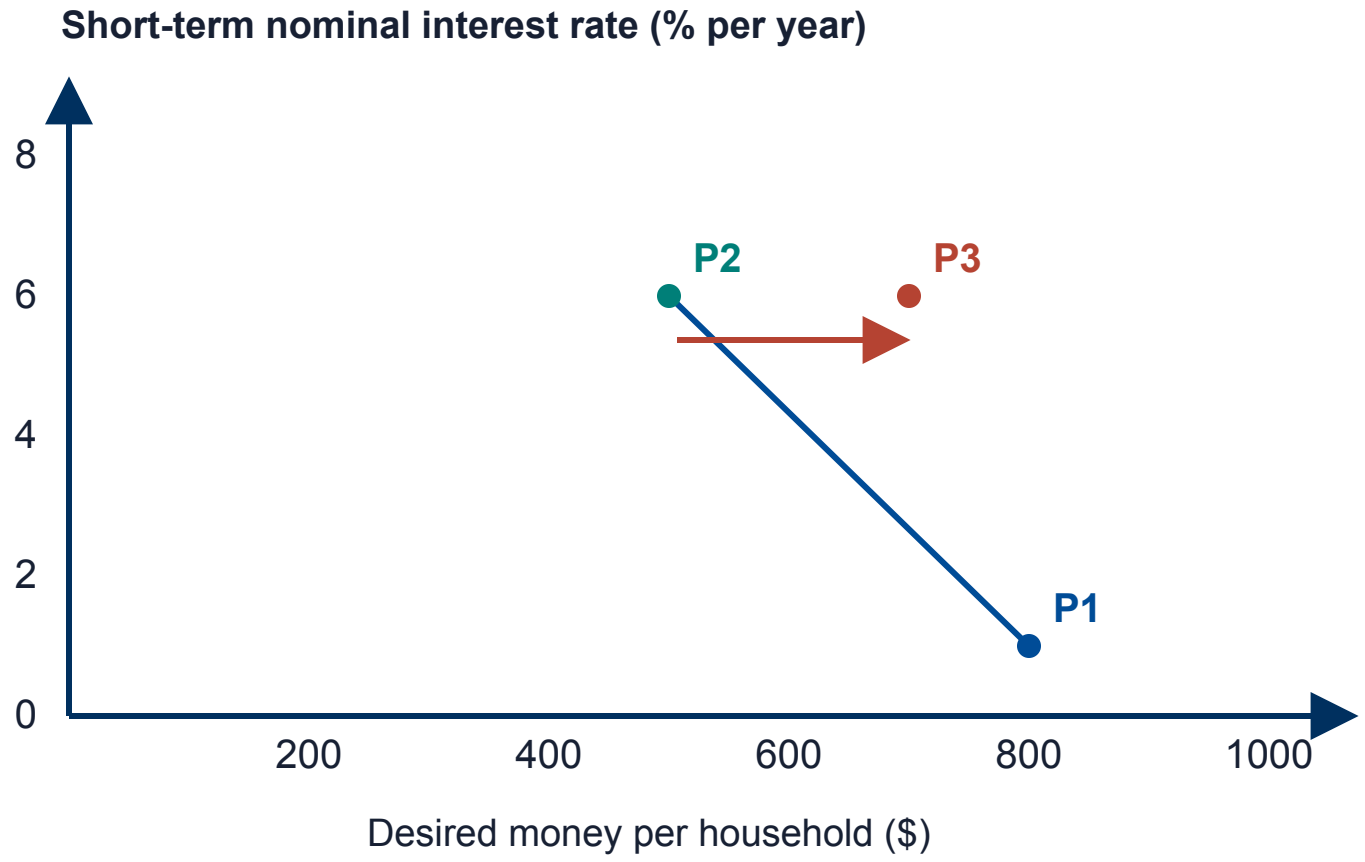
# The Opportunity Cost of Holding Money

| Holding $600 for one year | Ending balance | Interest forgone by holding money |
|---|---|---|
| **Cash/checking at 0%** | $600 | — |
| **One-year bond at 1%** | $606 | $6 |
| **One-year bond at 6%** | $636 | $36 |

- Rate of return = (ending value − initial value) ÷ initial value.
- A higher return on the alternative makes a liquid balance more costly to hold.
- Compare with the return paid on money itself; today the activity holds that return at 0%.

SEC Investor.gov: bonds · SEC Investor.gov: certificates of deposit

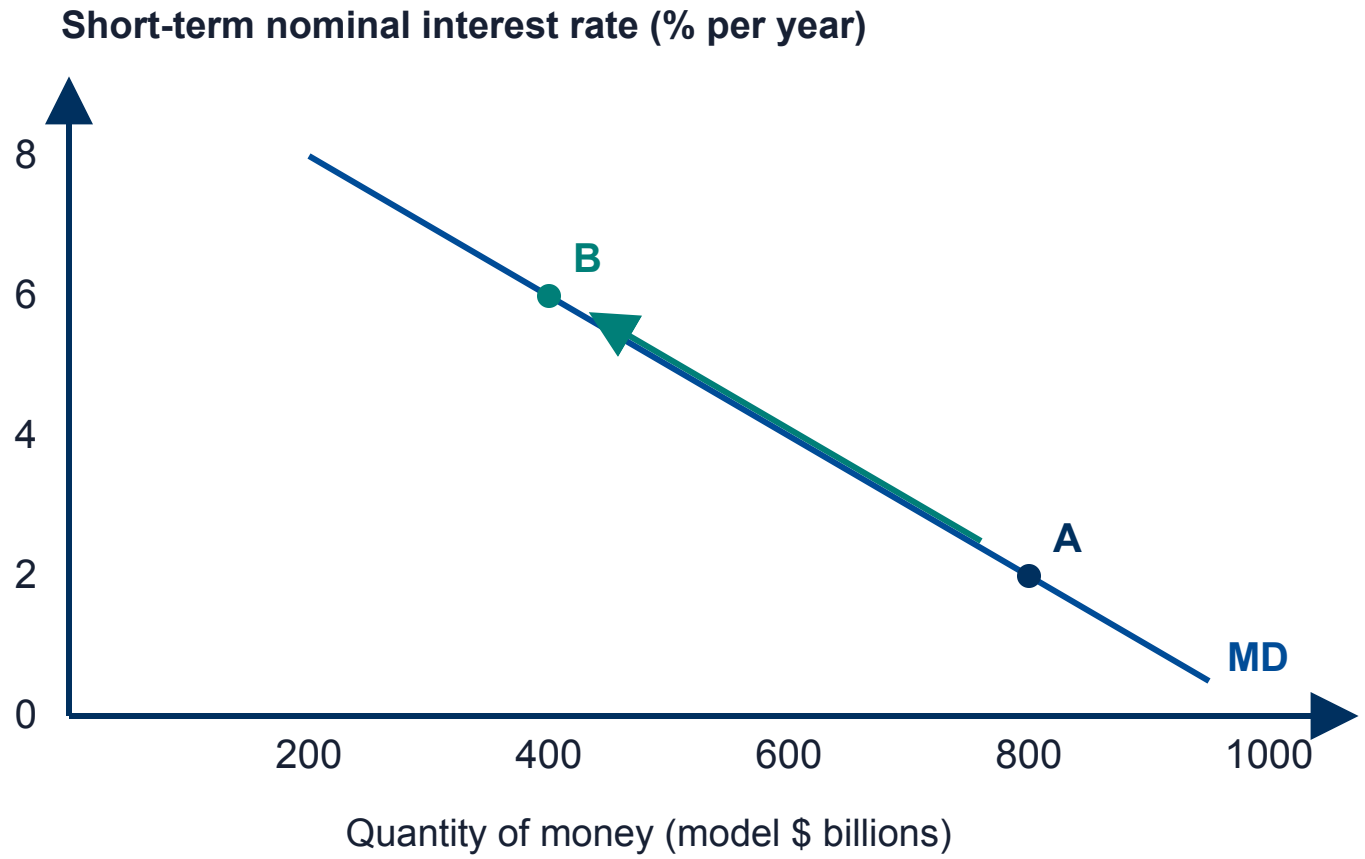
# From Choices to Money Demand




Schematic illustration; positions are model examples, not observed class responses.

1. P1 → P2: only the one-year return rises, from 1% to 6%.
2. At a higher short-term rate, the model predicts a smaller quantity of money demanded.
3. P2 → P3: bills become more expensive at the same 6% rate. That is a demand shift.

# The Demand for Money




Original short-run model: MD = 1,000 − 100i; fixed prices, real GDP, and money's own return.

1. Hold the price level, real GDP, payment arrangements, and money's own return fixed.
2. A rise in the short-term nominal interest rate raises the cost of holding money.
3. Move upward along MD: less money is desired for liquidity at the higher rate.

# Shifts in Money Demand

- Price level ↑ → the same real purchases require more dollars → MD shifts right.
- Real GDP ↑ → more real transactions at given prices → MD shifts right.
- Easier access to short-term credit can reduce the balances needed between paydays → MD may shift left.
- Faster, reliable banking and payments can reduce idle balances → MD may shift left.
- Institutions that require larger cash buffers can raise desired balances → MD shifts right.

# 15.2 Money and Interest Rates

- Explain the federal funds rate and the Fed’s influence on short-term rates.
- Interpret money-market equilibrium and shifts in the supply of money.
- Explain why longer-term rates reflect expectations and need not move one-for-one with today’s policy rate.

# The Federal Funds Rate

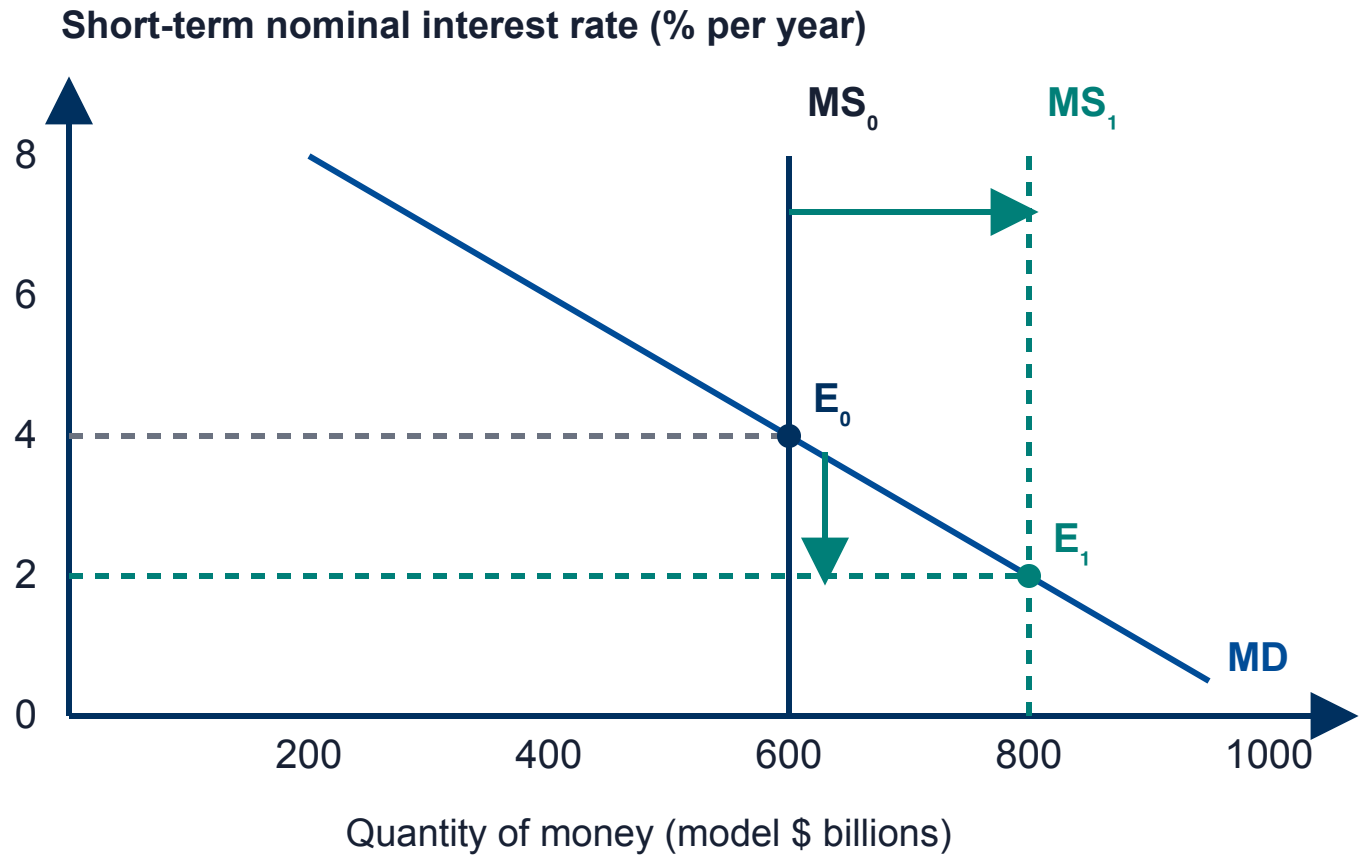
- Some banks lend reserve balances to other institutions overnight. The federal funds rate is the rate on those loans.
- The FOMC announces a target range; the market determines the effective rate.
- The Fed influences this overnight rate, which helps shape other short-term rates and broader financial conditions.

# From a Fed Purchase to Money Supply

1. Recall the previous class: the Fed buys a Treasury security from a nonbank seller.
2. The payment credits the seller’s bank with reserves; the bank credits the seller’s deposit.
3. The monetary base rises; the seller now holds a deposit instead of a security. In this model, MS shifts right.
4. The interest rate adjusts along the existing MD curve. Policy did not directly shift MD.

Today’s ample-reserves system relies mainly on administered rates such as IORB to steer overnight rates. A rate cut does not require a matching bond purchase.

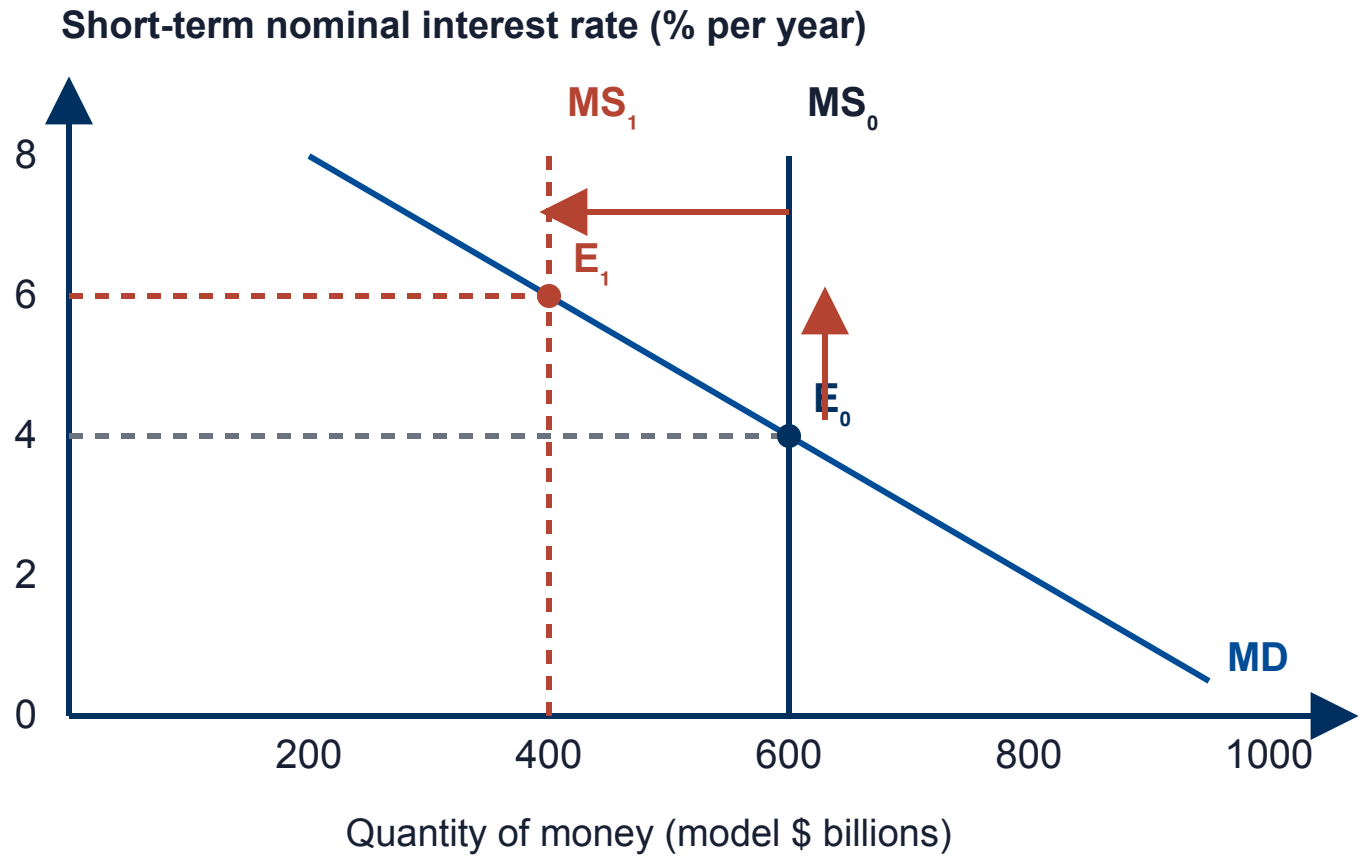
# Expansionary Monetary Policy




Original short-run model: MD = 1,000 − 100i; fixed prices, real GDP, and money's own return.

1. Start at $MS_0 = 600$ and $i_0 = 4\%$.
2. In the money-supply model, a purchase raises MS to 800.
3. At 4%, people hold more money than they desire; they seek bonds and other assets.
4. Bond prices rise and yields fall. New equilibrium: $i_1 = 2\%$, $M_1 = 800$.

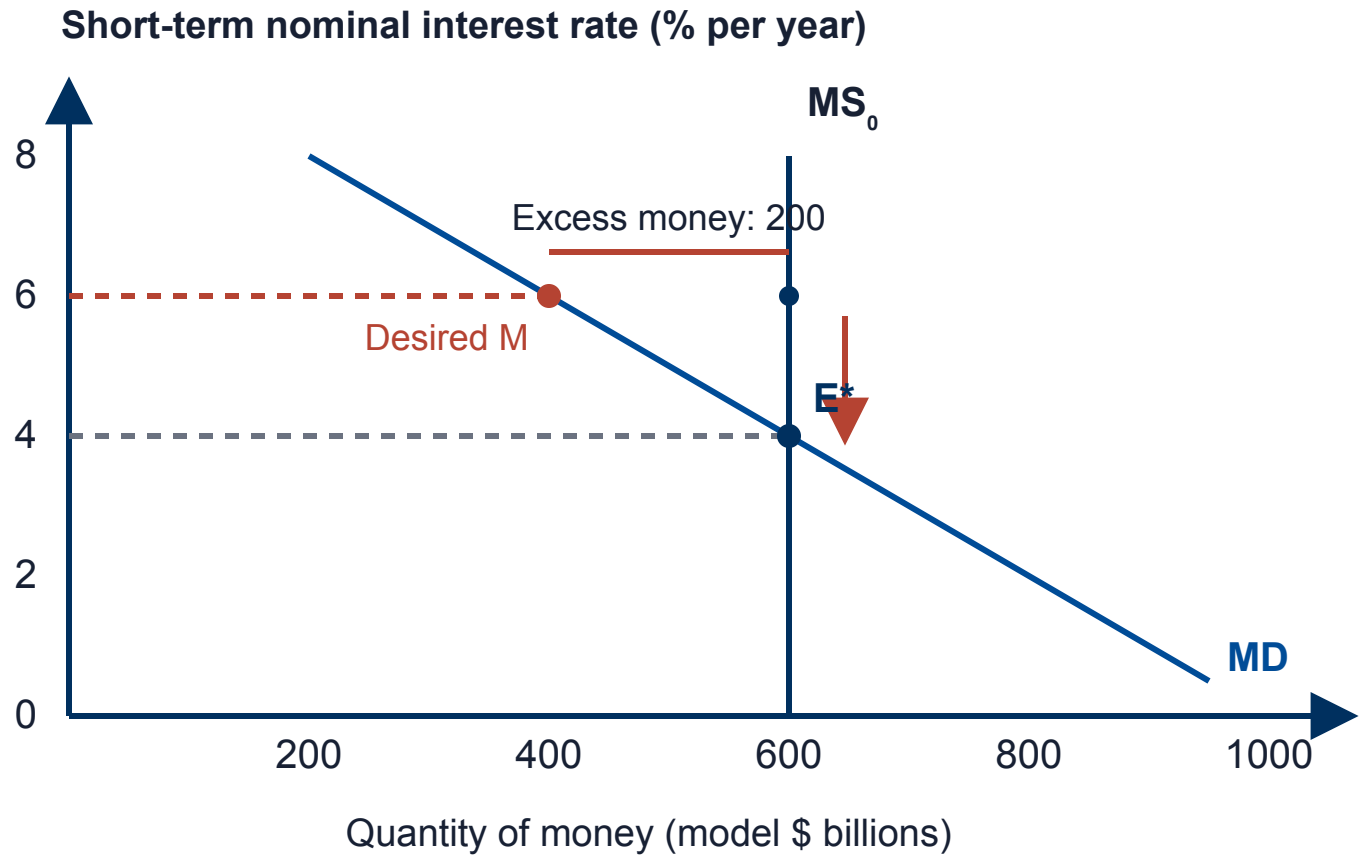
# Contractionary Monetary Policy




Original short-run model: MD = 1,000 − 100i; fixed prices, real GDP, and money's own return.

1. Start at $MS_0$ = 600 and $i_0$ = 4%.
2. In the money-supply model, a sale reduces MS to 400.
3. At 4%, people desire more money than is available; they sell assets to obtain balances.
4. Bond prices fall and yields rise. New equilibrium: $i_1$ = 6%, $M_1$ = 400.

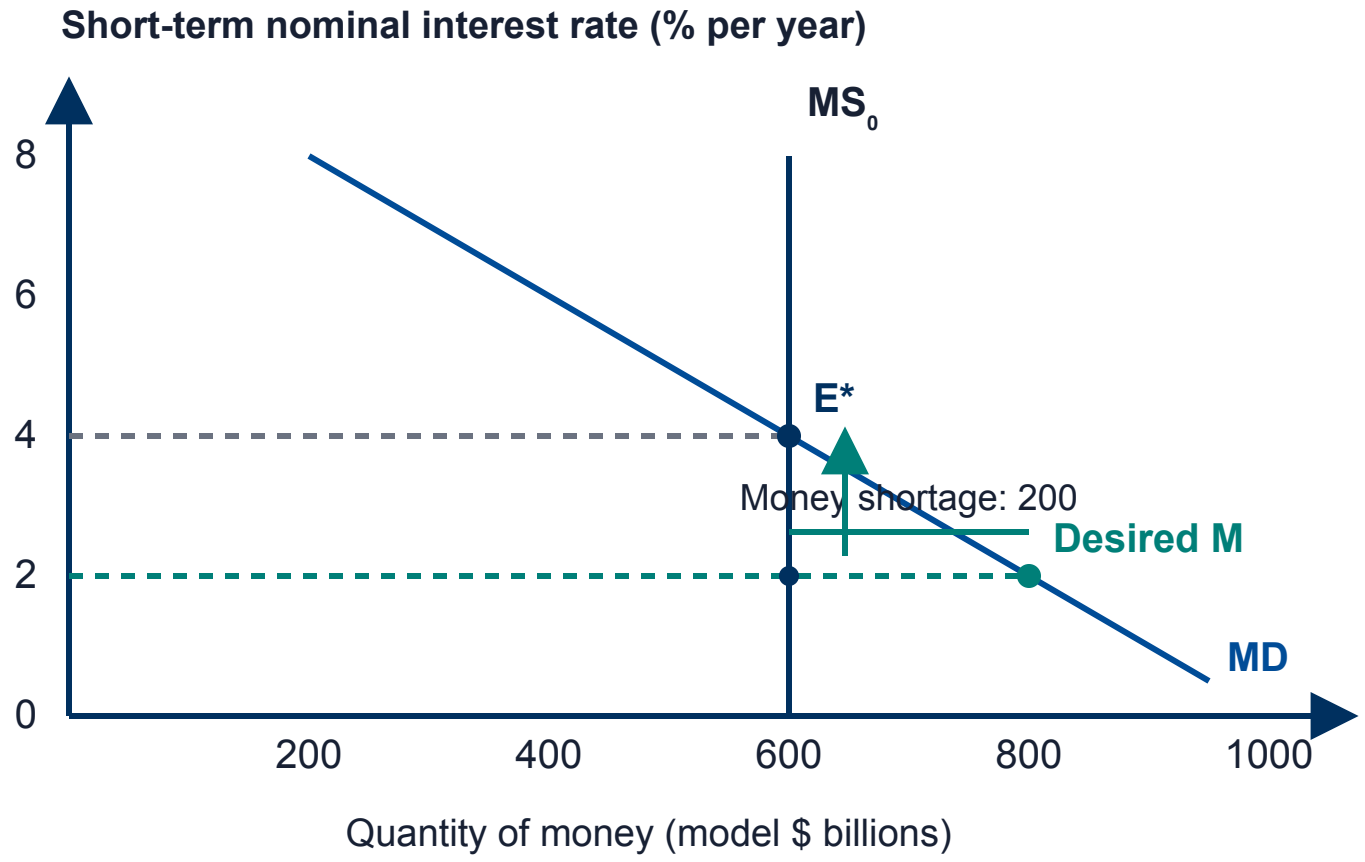
# Why a High Rate Does Not Persist




Original short-run model: MD = 1,000 − 100i; fixed prices, real GDP, and money's own return.

1. Suppose i = 6%: MD = 400 but MS = 600. Desired money holdings are below the available stock.
2. People try to exchange excess balances for bonds. The money remains in the system when it changes hands.
3. Bidding raises bond prices and lowers yields. A $100 one-year payoff rises from $94.34 to $96.15.
4. At i = 4%, desired and available money both equal 600: the adjustment pressure ends.

# Why a Low Rate Does Not Persist




Original short-run model: MD = 1,000 − 100i; fixed prices, real GDP, and money's own return.

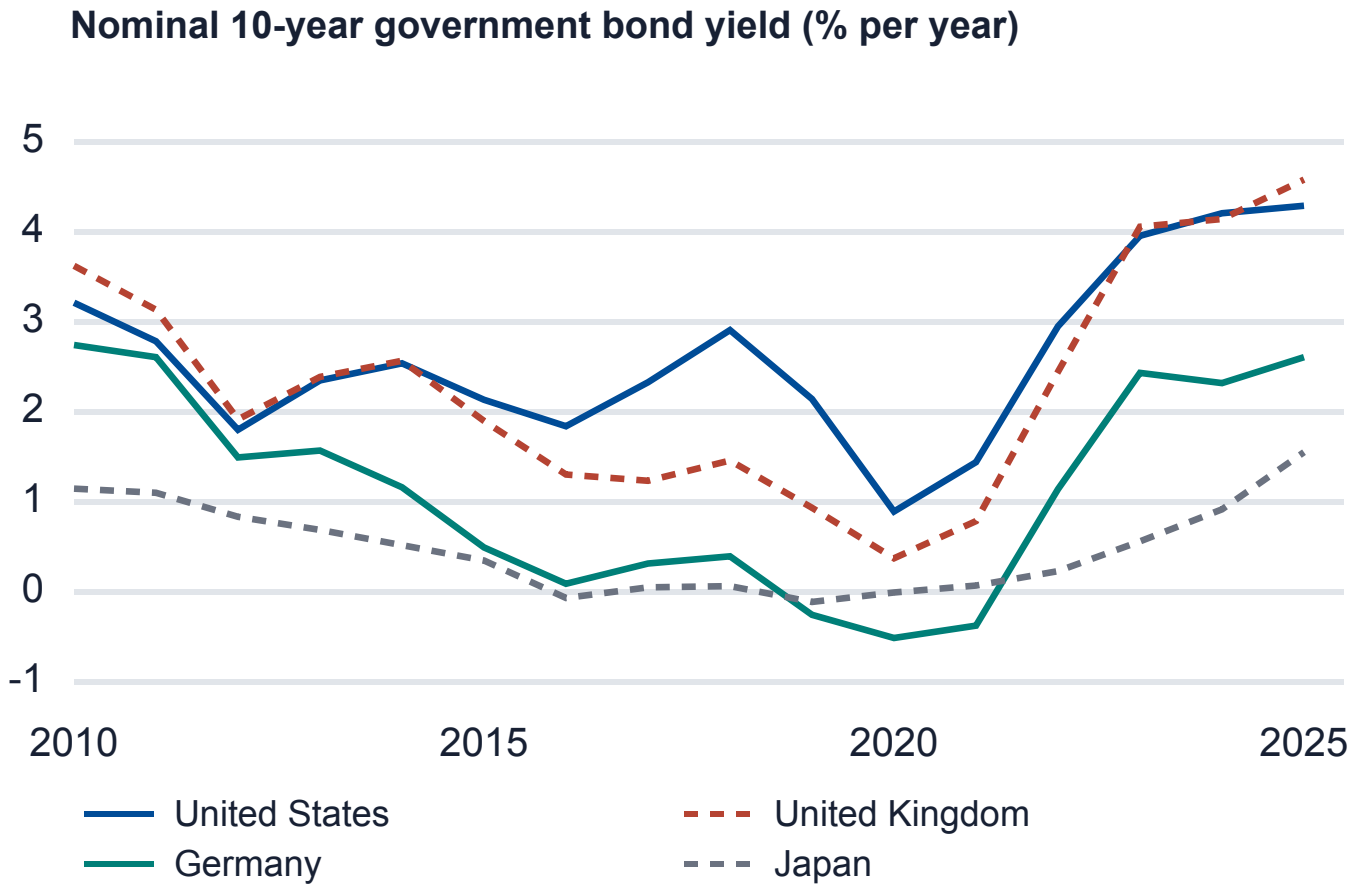
1. Suppose i = 2%: MD = 800 but MS = 600. People want more liquid balances.
2. Households sell bonds to build balances for rent and bills; not everyone can raise money holdings at once.
3. Bond prices fall and yields rise. A $100 one-year payoff falls from $98.04 to $96.15.
4. At i = 4%, desired and available money both equal 600.

# Money-Market Effects

| Policy in the model | Money supply | Bond price | Short-term rate |
|---|---|---|---|
| **Expansionary purchase** | Increases | Rises | Falls |
| **Contractionary sale** | Decreases | Falls | Rises |

Hold money demand, the price level, real GDP, and money’s own return fixed.

# Long-Term Interest Rates across Countries




Annual averages, 2010–2025; own-currency yields. Original chart from OECD data.

1. Read first: annual-average yields on approximately 10-year government bonds, in each country's currency.
2. Compare Japan and the United States: their yields differ despite the same broad maturity.
3. The rates rise after 2021 in all four series, by different amounts. The chart alone does not identify the cause.

OECD (2026), Financial market statistics: long-term interest rates (accessed November 2, 2026) · OECD data terms, section 3

# One Year or Two Years?

| Invest $600 for two years | Year 1 rate | Year 2 rate |
|---|---|---|
| **Two-year contract, annual compounding** | 5% fixed | 5% fixed |
| **Roll one-year investments: P5** | 4% | 2% scenario |
| **Roll one-year investments: P6** | 4% | 8% scenario |

- Answer PollEv P5 and P6 before the payoff appears.
- P5: $\$600 \times 1.05^2 = \$661.50$; rolling gives \$636.48. The two-year contract pays more.
- P6: rolling gives $\$600 \times 1.04 \times 1.08 = \$673.92$. The one-year strategy pays more.

# Why Maturity Matters

- A longer-term yield reflects the expected path of short-term rates and compensation for holding longer-term risk.
- A change in today's policy rate may already be anticipated; a 10-year yield need not change by the same amount.
- Longer maturity is different from the macroeconomic long run. A two-year investment is longer than a one-year investment; the data chart uses a 10-year benchmark.

SEC Investor.gov: bonds · Federal Reserve Board: policy goals and transmission

# 15.3 Monetary Policy and Aggregate Demand

- Explain how monetary policy affects consumption, investment, and aggregate demand.
- Interpret policy-induced AD shifts while holding short-run aggregate supply fixed.
- Demonstrate expansionary and contractionary effects on output and the price level.

# Fiscal and Monetary Policy

| Policy choice | Main route | When especially useful |
| --- | --- | --- |
| **Monetary policy** | Changes financial conditions; affects C and I | Broad demand management with room to adjust rates |
| **Fiscal policy** | Government purchases, taxes, and transfers | Direct purchases or support targeted to affected groups |
| **A coordinated response** | Combines instruments | Severe downturns, especially near the effective lower bound |

- Automatic stabilizers can act without a new law; discretionary fiscal measures may take time to enact.
- The choice also depends on inflation, implementation delays, debt capacity, and the type of shock. A supply loss is not repaired simply by adding demand.

IMF: fiscal policy · Federal Reserve Board: policy goals and transmission

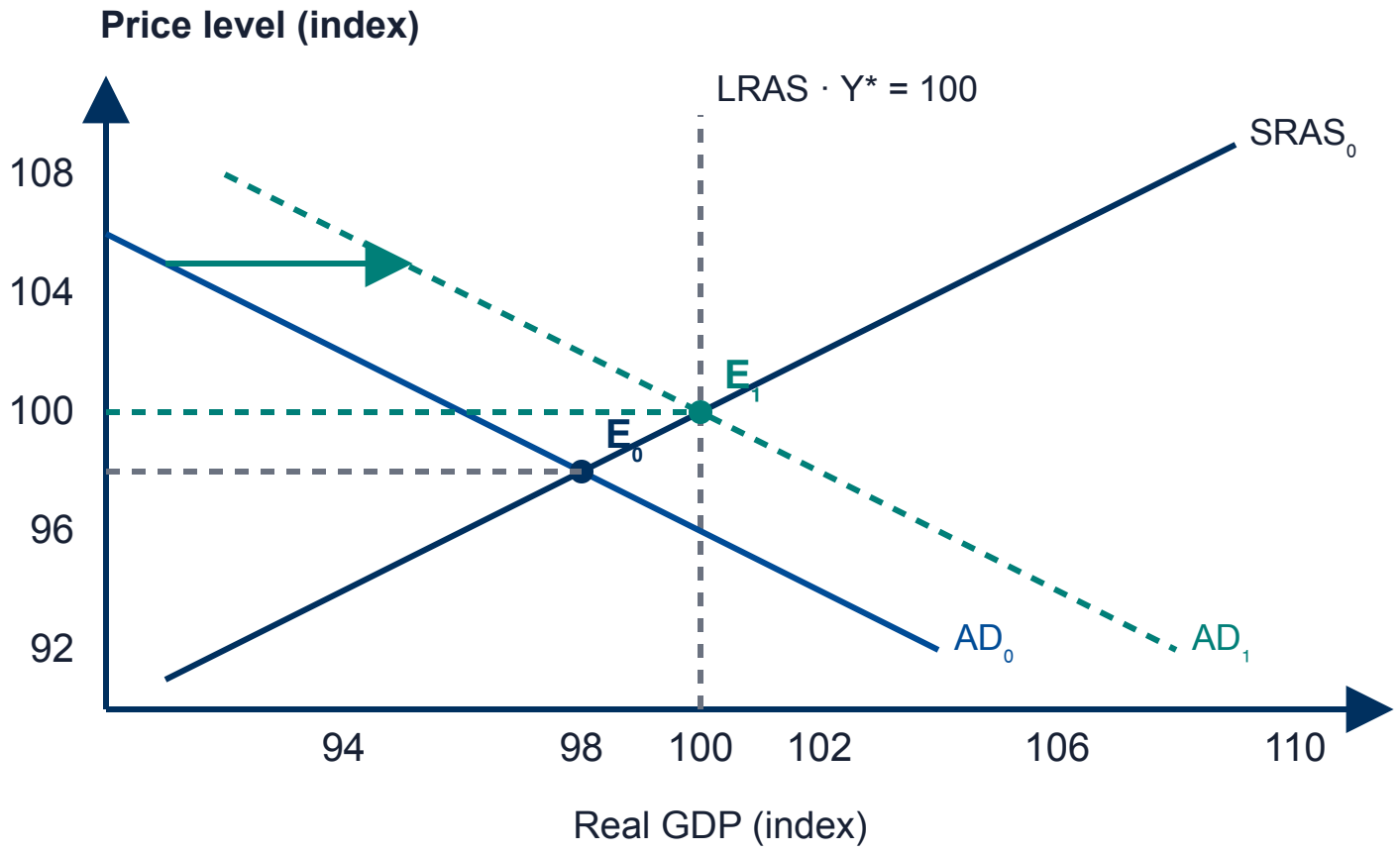
# From Interest Rates to Aggregate Demand

1. A lower policy rate can ease broader borrowing and financial conditions.
2. Lower relevant real borrowing costs can encourage consumption and investment.
3. A $10,000 project paying $10,500 next year loses money with a $10,600 repayment but gains with a $10,400 repayment.
4. At a given price level, higher C and I raise planned expenditure: AD shifts right.

Transmission is conditional: expected inflation, credit access, risk, and confidence also matter.

Federal Reserve Board: policy goals and transmission

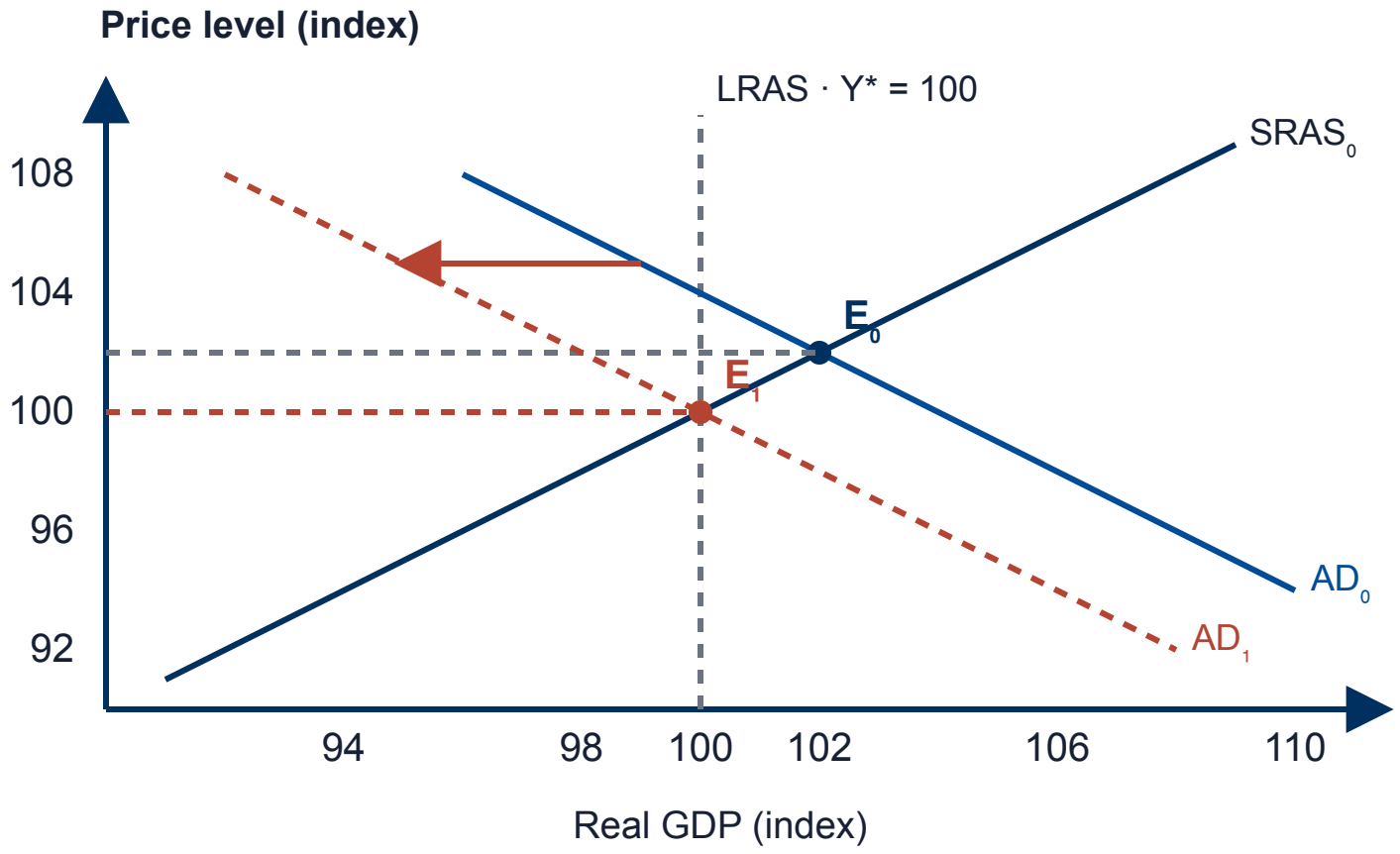
# Expansionary Policy and Aggregate Demand




Original illustrative model; SRAS and potential output stay fixed during the AD policy comparison.

1. Begin below potential output: $Y_0 = 98$, while $Y^* = 100$.
2. Easier financial conditions encourage interest-sensitive C and I.
3. AD shifts right, from $AD_0$ to $AD_1$; SRAS and LRAS stay fixed.
4. In this short-run model, output and the price level rise to 100.

# Contractionary Policy and Aggregate Demand




Original illustrative model; SRAS and potential output stay fixed during the AD policy comparison.

1. Begin above potential output: $Y_0 = 102$, while $Y^* = 100$.
2. Tighter financial conditions restrain interest-sensitive C and I.
3. AD shifts left, from $AD_0$ to $AD_1$; SRAS and LRAS stay fixed.
4. In this short-run model, output and the price level are lower than without the tightening.

# Check the Three Objectives

- A · At a fixed 6% rate, the same bill basket becomes more expensive. Movement or shift in money demand?
- B · With MD fixed, money supply increases. What happens to the equilibrium short-term rate?
- C · Trace a rate decrease through C and I to AD, output, and the price level.

**Explanation.** A: MD shifts right. B: the equilibrium rate falls. C: if financial conditions ease, C and I rise, AD shifts right, and short-run output and the price level rise with SRAS fixed.